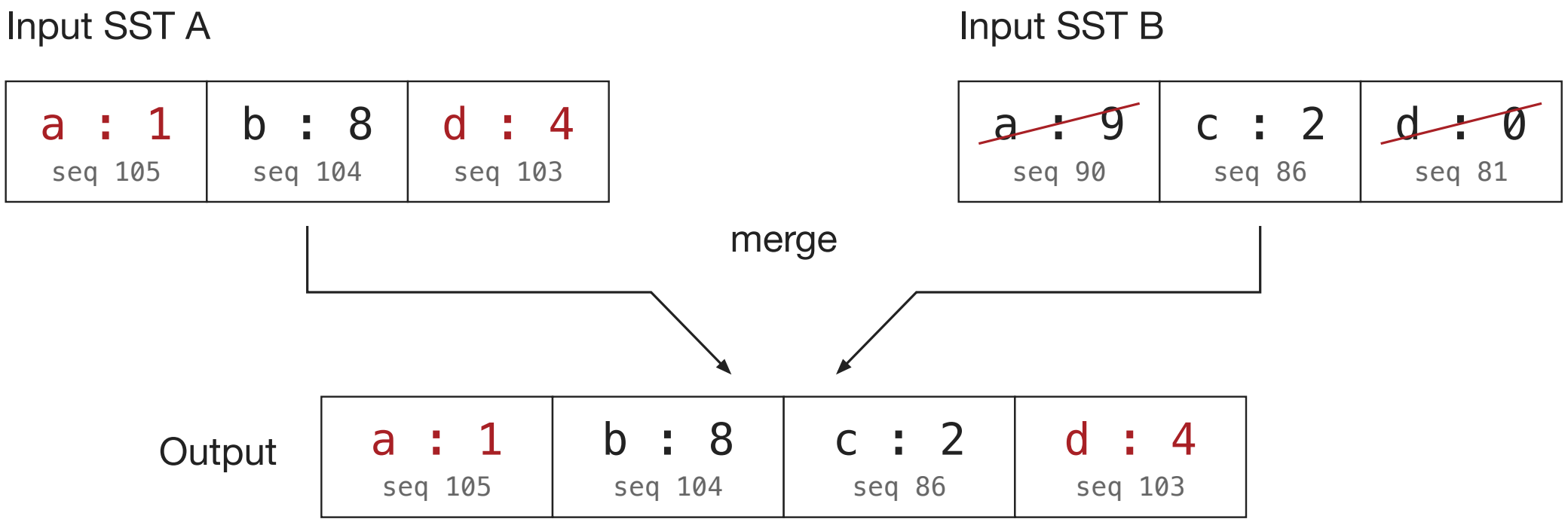
Compaction is a sorted merge

Textbook theme · precisely placed record diagram · CeTZ

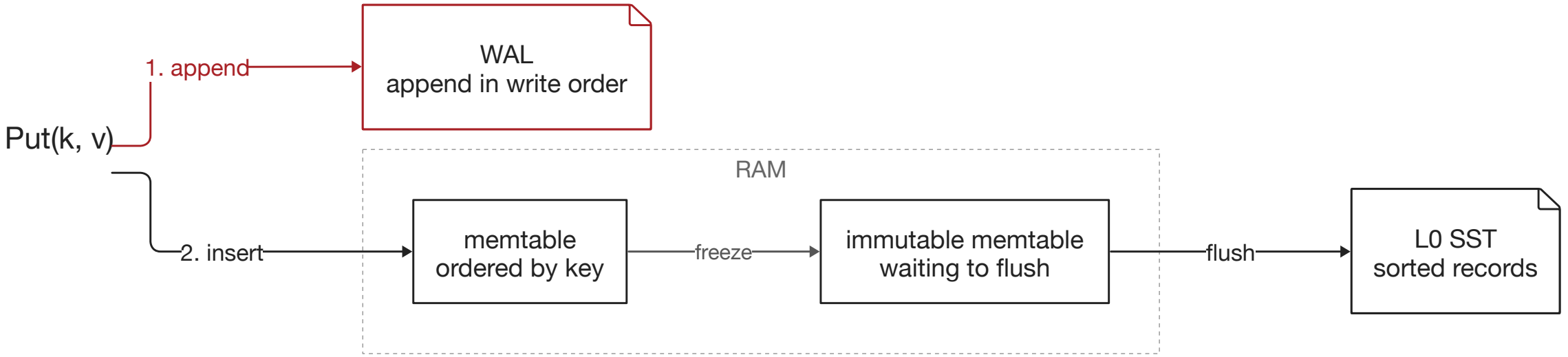


Here, the larger sequence number wins for each key.

Illustrative inputs. Older versions shown here have no snapshot that needs them. Output remains sorted by key.

Put writes a log and an ordered buffer

Textbook theme · automatic layout · D2, clean rendering

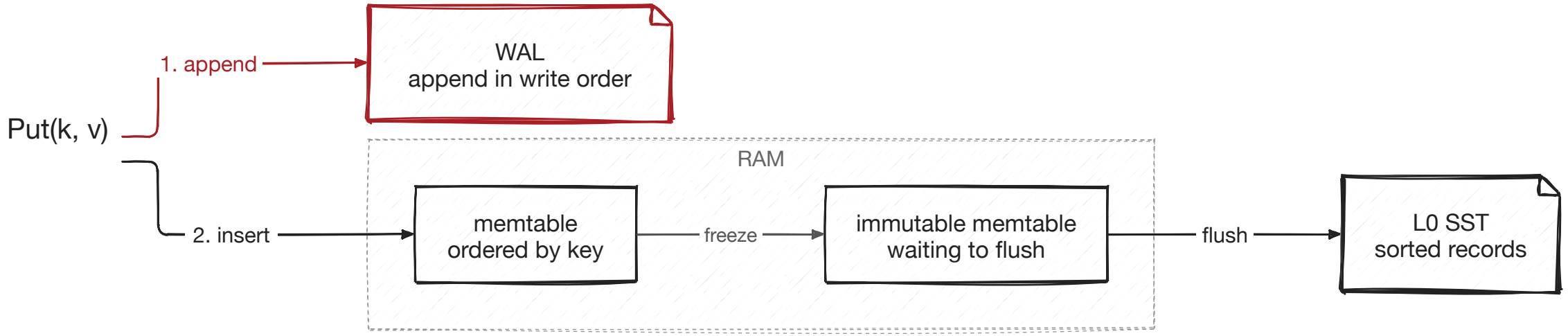


The WAL preserves write order; the SST preserves key order.

Ordinary WAL-enabled path. A synchronous write also waits for WAL sync; background flush creates an L0 SST.

Put writes a log and an ordered buffer

Textbook theme · same D2 source · sketch rendering



The WAL preserves write order; the SST preserves key order.

Ordinary WAL-enabled path. A synchronous write also waits for WAL sync; background flush creates an L0 SST.